MIGUEL VILÁ GONZÁLEZ

miguel-vila @ github

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Remote - US East Coast Timezone
SiriusXM — *November 2022 - Present*

- Member of the API tooling team, part of the platform services enablement organization.
- Develop tooling using Smithy, an AWS DDL language, empowering teams to describe, implement, and consume their services.
- Key projects include:
  - Developed a system for describing which Smithy services are implemented and consumed by applications, enabling package management and dependency tracing across the platform.
  - Implemented compatibility checks to verify service changes against potential data processing breakages.

- Contributed to a team implementing semantic search using vector embeddings stored in OpenSearch.
  - Set up infrastructure to ingest catalog data and generate embeddings using different models.
  - Set up a client to query the different indexes using the embeddings.

- Led a small team implementing a reusable ingester, with the objective of allowing teams to build tables from CMS data.
  - Led conversations in order to define the requirements and solve ambiguities.
  - Proposed and implemented a solution using Scala and AWS services (a Lambda consuming a Kinesis stream, a target DynamoDB table).

- As part of the Playback team, contributed:
  - Collaborated with other teams in order to deliver a new feature allowing users to listen to stations generated from their liked content.
  - Proposed and designed changes to the e2e tests and smoke tests. Delegated implementation tasks to team members.

Lead Software Engineer ← Senior Software Engineer ← Software Engineer — United Kingdom
Disney (formerly Disney Streaming Services) — *January 2017 - October 2022*

- Built and maintained Disney+ and ESPN+ streaming platforms as a member of various engineering teams, utilizing Scala, Python, TypeScript, and AWS services.
- Developed and maintained commerce operation services, including subscription lifecycle management, collaborating with cross-functional teams on feature delivery.
- Participated in the evolution, migration, and unification of the subscription system.

- Led a project migrating one of the subscription system's clients to a next-generation version:
  - Crafted technical proposals and guided decision-making.
  - Collaborated with client teams through technical discussions to ensure thorough understanding of requirements and edge cases.
  - Conducted research to inform the implementation.
  - Improved documentation to streamline onboarding.
  - Decomposed project work into discrete, deliverable units.

- Other responsibilities and achievements included:
  - Designed and proposed new components to enhance the subscription management system's API and domain events, collaborating with engineers on design and behavior. Resulted in improved data quality for analytics and increased visibility into the subscription lifecycle.
  - Developed and maintained a component to detect subscription overlaps between partner-issued and existing user subscriptions, automatically applying pauses or discounts per business rules. This enhanced user experience and reduced customer service costs.

- Onboarded new team members to the subscription team, providing domain primers and iteratively improving internal documentation.
- Authored multiple technical proposals, presenting various options, advocating for preferred solutions, and decomposing them into actionable tasks for team execution.
- Proposed and implemented a change to test-subscription creation logic, removing reliance on naming conventions. This enabled complex scenario testing for other teams, simplified test data identification and cleanup, and reduced storage costs.

- Developed user authentication, account management, and profile management services, including multiple Scala-based microservices, as part of a separate team.
  - Executed performance tests against these services to define appropriate scaling policies.
  - Helped establish the infrastructure for multiple projects.

Software Engineer Colombia
s4n *January 2013 - June 2016*

- Led a small engineering team, responsible for technical decision-making and mentoring new members on the tech stack (Java, Scala, JavaScript).
- Developed enterprise software for diverse clients, primarily in the insurance sector, using Java, Scala, and JavaScript.
- Developed a REST API using Scala for an international client in the United States.

Teaching Assistant for Design and Analysis of Algorithms Colombia
Los Andes University *August 2011 - May 2012*

- Evaluated student homework and exams. Organized and delivered preparatory lectures for exams.

Software Engineer Colombia
CIACUA, Los Andes University *March 2011 - August 2011*

- Participated in the software reengineering of an application used for the design and simulation of water distribution systems. Documented the design in UML and fixed bugs found in a version update.

EDUCATION

Recurse Center *August 2016 - November 2016*
United States of America

Completed a self-directed, project-based residency for programmers, focusing on deepening expertise outside core specializations. Developed projects focused on distributed systems, concurrency, and functional programming.

Los Andes University *January 2009 - December 2012*
Colombia

BSc in Systems and Computing Engineering

COMPLEMENTARY EDUCATION

- Digital Product Management: Modern Fundamentals, University of Virginia (via Coursera). Completed 2026.
- Building Agentic AI Applications with a Problem-First Approach (via Maven). Completed 2025.
- AI Builders Bootcamp (via Maven). Completed 2025.
- Senior Engineer to Lead: Grow and thrive in the role (via Maven). Completed 2025.
- Cloud Computing Applications, Part 1: Cloud Systems and Infrastructure, University of Illinois Urbana-Champaign (via Coursera). Completed 2025.
- AI Agents Fundamentals, Hugging Face. Completed 2025.
- A Technical Leader's Qualities and Effectiveness, University of Colorado Boulder (via Coursera). Completed 2024.

- Introduction to Corporate Finance, University of Pennsylvania (via Coursera). Completed 2024.
- Financial Markets (with Honors), Yale University (via Coursera). Completed 2021.
- Intro to Data Science, Washington University (via Coursera). Completed 2013.
- Machine Learning, Stanford University (via Coursera). Completed 2012.

ACHIEVEMENTS AND AWARDS

- *"Quiero Estudiar"* Scholarship. Awarded full scholarship for undergraduate studies at Los Andes University (October 2008).
- Qualified for the Latin American Regional International Collegiate Programming Contest (ICPC), 2011.
- Qualified for the Latin American Regional International Collegiate Programming Contest (ICPC), 2012.

TECHNICAL SKILLS

Programming Languages (in order of skill):	Scala, Java, Python, Javascript, TypeScript.
Infrastructure:	Mostly AWS. Cloudformation, CDK, DynamoDB, Kinesis, ECS, CloudWatch.

TALKS

- Más allá de los contratos: Platform Engineering con Smithy - DevOpsDays 2026

OPEN SOURCE SOFTWARE CONTRIBUTIONS

Over the course of my career, I have contributed to numerous open-source projects. These contributions were often driven by project requirements—fixing bugs or adding features necessary for my work. Below are some of my most notable contributions:

- **orca #298**: Added an effort setting (`--effort`) for Claude agent invocations, allowing users to tune the cost/latency vs. quality tradeoff per run.
- **opensearch-api-specification #324**: Enhanced the OpenSearch API specification testing framework to allow step output chaining, enabling more complex test scenario execution.
- **opensearch-java #1005** & **opensearch-api-specification #304**: Extended the OpenSearch Java client to support text embeddings.
- **smithy #2221**: Extended OpenAPI conversion to correctly handle deprecated annotations.
- **spotbugs #2988**: Fixed a bug in the SpotBugs static analysis tool related to reference comparison linting, addressing a common source of Java errors.
- **smithy-translate #42**: Fixed a critical bug in a dependency function, addressing an edge case unknown to maintainers and restoring library utility.
- **zio-metrics #53**: Improved the external API of a metrics library for enhanced usability.
- **scalaz #750**: Added new functionality, applying learned functional programming concepts.

SIDE PROJECTS

Personal projects available at: invariante.co/side-projects.html